

Quantum Mechanics – Homework 2

Name: _____ Due Date: 2026.09.23

This homework covers Lectures 4–5. Show the main steps in every calculation. All wave functions must be normalized before they are used to calculate probabilities or expectation values. You may use

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}, \quad n = 1, 2, 3, \dots$$

Problem 1. Normalization and the Born Rule (20 points)

A particle in one dimension is described by

$$\psi(x) = \begin{cases} Ae^{-\kappa|x|}, & -\infty < x < \infty, \quad \kappa > 0, \end{cases}$$

where A is real and positive.

- (a) State the normalization condition for this wave function. Use the symmetry of $|\psi(x)|^2$ to rewrite the integral over the whole real line as an integral over $x > 0$. (4 points)
- (b) Determine the normalization constant A . (6 points)
- (c) Find the probability of detecting the particle in the interval $-1/\kappa < x < 1/\kappa$. Give both the exact result and a decimal value. (6 points)
- (d) Determine the physical dimensions of A and explain why $|\psi(x)|^2$ is a probability density rather than a probability. (4 points)

Problem 2. Normalization of a Superposition (20 points)

On the interval $0 < x < L$, define

$$\phi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right), \quad \phi_2(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right).$$

Both functions vanish outside the interval. Consider the superposition

$$\psi_\alpha(x) = N [\phi_1(x) + e^{i\alpha} \phi_2(x)],$$

where α is real and N is real and positive.

Reminder from Lecture 4. Recall the following question from class. Consider a sine wave of finite length L :

$$f(x) = \begin{cases} \sin(kx), & -\frac{L}{2} < x < +\frac{L}{2}, \\ 0, & \text{elsewhere.} \end{cases}$$

Which statement is closest to the truth?

- A) $f(x)$ has a single well-defined wavelength.
- B) $f(x)$ is made up of a range of wavelengths.

Answer: B. This is an example of the same point used in the present problem. A sine wave restricted to a finite interval is not a plane wave extending over all space. Its cutoffs are part of the complete function, so its Fourier expansion contains a range of wavelengths and momenta. Therefore ϕ_1 and ϕ_2 should be treated as functions defined on $0 < x < L$, and neither function has one definite momentum merely because a sine appears in its definition.

- (a) Verify that ϕ_1 and ϕ_2 are individually normalized and show that

$$\int_0^L \phi_1(x) \phi_2(x) dx = 0.$$

(5 points)

- (b) Normalize ψ_α . Show explicitly why N does not depend on the relative phase α . (5 points)

- (c) Show that the probability density is

$$|\psi_\alpha(x)|^2 = \frac{1}{2} [\phi_1^2(x) + \phi_2^2(x) + 2 \cos \alpha \phi_1(x) \phi_2(x)].$$

Identify the interference term.

(4 points)

(d) Calculate $P(0 < x < L/2)$ and show that

$$P(0 < x < L/2) = \frac{1}{2} + \frac{4 \cos \alpha}{3\pi}.$$

Evaluate this probability for $\alpha = 0$, $\pi/2$, and π .

(6 points)

Problem 3. Position Statistics (20 points)

A particle is described by

$$\psi(x) = \begin{cases} Ax(L-x)^2, & 0 < x < L, \\ 0, & \text{otherwise,} \end{cases}$$

where A is real and positive.

- (a) Normalize the wave function and determine A . (4 points)
- (b) Calculate the probability of finding the particle in the left half of the interval, $0 < x < L/2$. (3 points)
- (c) Calculate $\langle x \rangle$ and $\langle x^2 \rangle$. (7 points)
- (d) Calculate the position uncertainty

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}.$$

(4 points)

- (e) Find the most probable position by locating the maximum of $|\psi(x)|^2$. Is it equal to $\langle x \rangle$? Explain the distinction between the most probable value and the expectation value. (2 points)

Problem 4. The Momentum Operator (20 points)

On $0 < x < L$, consider the normalized wave function

$$\psi_q(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) e^{iqx},$$

with $\psi_q(x) = 0$ outside the interval. Here q is real. Use

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}.$$

- (a) Explain why multiplication by e^{iqx} does not change the position probability density or the normalization. (3 points)
- (b) Calculate $\hat{p}\psi_q(x)$. Is ψ_q an eigenfunction of momentum? Explain. (4 points)
- (c) Calculate the expectation value $\langle p \rangle$. (5 points)
- (d) Calculate $\langle p^2 \rangle$ and the momentum uncertainty Δp . (6 points)
- (e) State how $\langle p \rangle$ and Δp change when q is increased. (2 points)

Problem 5. Momentum-Space Statistics (20 points)

Consider a particle on a periodic interval $0 \leq x \leq L$. Let

$$k = \frac{2\pi}{L}, \quad u_+(x) = \frac{1}{\sqrt{L}}e^{ikx}, \quad u_-(x) = \frac{1}{\sqrt{L}}e^{-ikx},$$

and

$$\psi(x) = \frac{\sqrt{3}}{2}u_+(x) + \frac{i}{2}u_-(x).$$

- (a) Show that u_+ and u_- are normalized and orthogonal on $0 \leq x \leq L$. Hence verify that ψ is normalized. (5 points)
- (b) Apply $\hat{p} = -i\hbar\partial_x$ to u_+ and u_- . List the two possible results of a momentum measurement and their probabilities. (4 points)
- (c) Calculate $\langle p \rangle$, $\langle p^2 \rangle$, and Δp . (7 points)
- (d) Calculate $|\psi(x)|^2$. Explain why a state made from two definite-momentum plane waves can have a position-dependent probability density. (4 points)

End of Homework 2